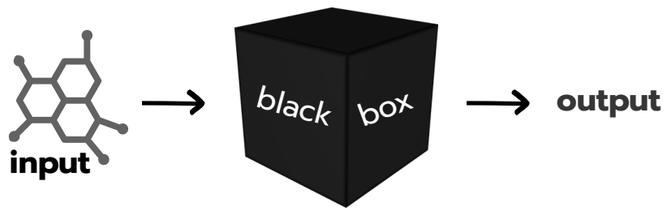
Cursos Iniciación a la Investigación 2026
Vicerrectoría de Investigación e Innovación

LLMs as an Interpretative Layer in XAI Pipelines

What happens when an LLM mediates the explanation of a machine learning model?

Student: Manuela Guédez - Professor: Fiorella Cravero

1. INTRODUCTION



Black - Box Problem

- Modern Machine learning (ML) models operate as black boxes, creating a gap between predictive power and human understanding [1].
- eXplainable Artificial Intelligence (XAI) methods provide technically correct explanations
  - But their outputs remain cognitively inaccessible to non-experts.
- Large Language Models (LLMs) can translate these outputs into natural language.
  - However, LLMs may alter, simplify, or hallucinate explanatory content. [2, 3].

This work proposes a modular pipeline and formal metrics to analyze LLMs as interpretative agents, in XAI systems.

Hypotheses:

H1: LLMs improve human interpretability by acting as semantic interpretive interfaces over XAI outputs

H2: Prompting strategies modulate epistemic risk, including hallucination, distortion, and structural misalignment.

2. METHODOLOGY

We design a modular pipeline in which a predictive model is explained using post-hoc XAI methods. Then tructured explanations are transformed into controlled prompts and processed by an LLM acting as an interpretative layer. The resulting explanations are formally evaluated in terms of fidelity and factuality.

Cheminformatics domain

- **TIRESIA**[4]: Toxicity
- **BCF** [5]: Bioconcentration Factor

Supervised learning models

- **TIRESIA**: Random Forest Classifier (RFC)
- **BCF**: RFC, XGBoost (XGB).

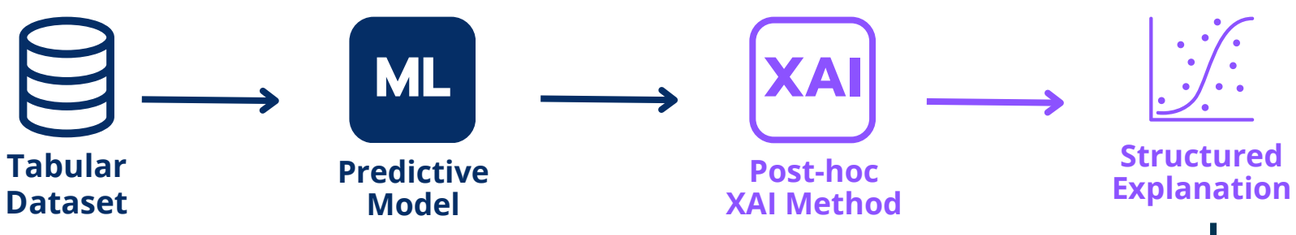
Model-agnostic explanation techniques

- **SHAP** [6]
- **LIME** [7]

Extract feature-attribution information from ML model.

Output XAI

- Feature Ranking
- Contribution signs
- Contribution magnitudes



Explanation Quality Assessment

- **FIDELILY [Fid]:** a composite index of ranking [F_r] alignment (Spearman over top-k features), directional consistency [F_d], and relational structure [F_l] preservation.

  Fid = $\beta_1 \cdot F_r + \beta_2 \cdot F_d + \beta_3 \cdot F_l$ (equal weights).

- **FACTUALITY [Fac]:** the alignment between LLM-generated statements [N_t] and XAI outputs. Statements are classified as correct [N_c], partially correct [N_p], or non-factual (hallucinated).

  Fac = $(N_c + \alpha \cdot N_p) / N_t$, with $\alpha = 0.5$.

Evaluation

Narrative Explanation

LLM-mediated Explanation

Semantic mediation (not new model explanation)

Interpretative Layer

Prompt Engineering [8]:

- Context awareness
- Expertise adaptation
- Actionability
- Step-by-step reasoning
- Explicit constraints on raw input reproduction

LLM

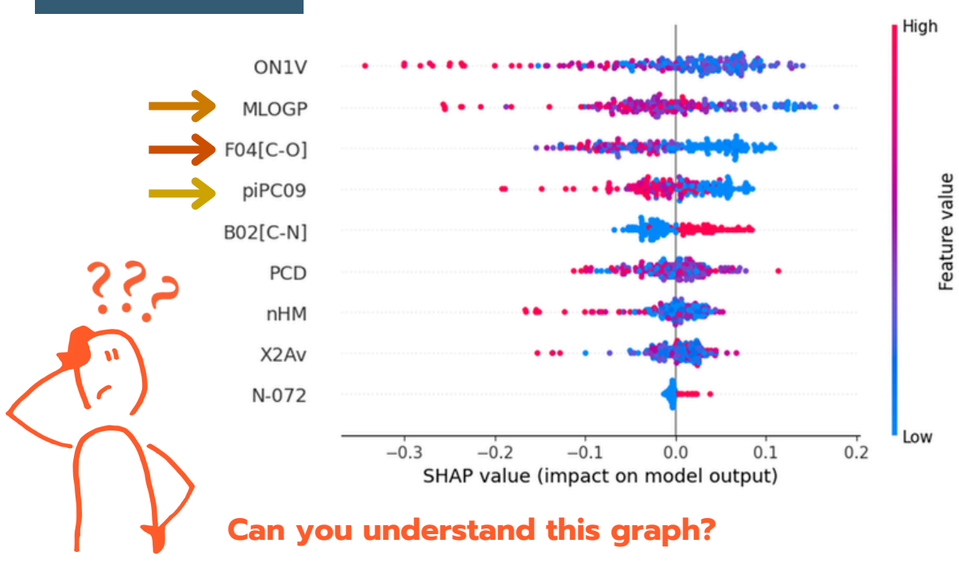
Gemini 2.5 Flash [9]

3. RESULTS AND DISCUSSION

3.1. ML RESULTS

ML model	Accuracy	F1 - score	Recall
RFC (BCF)	0.69	0.77	0.81
XGB (BCF)	0.67	0.76	0.81
RFC (Tiresia)	0.77	0.74	0.67

3.2. XAI OUTPUT



Can you understand this graph?

3.3. LLM OUTPUT

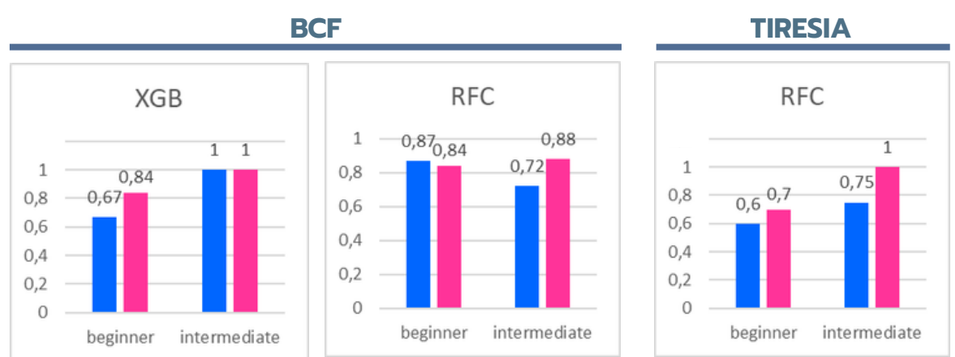
Top three most influential molecular descriptors (with direction of influence):

- *F04[C–O] (absent) → Negative contribution toward high bioconcentration (pushes prediction away from Class 3).*
- *MLOGP (6.47) → negative contribution toward bioconcentration (lipophilicity effect).*
- *piPC09 (6.609) → Negative contribution toward high bioconcentration (higher polarity reduces accumulation).*

What about this explanation? Isn't it much better?

- However, **a good explanation is not necessary a true one**.
- To asses whether mediation preserves explanatory truth, **formal evaluation is required**.

3.4. EXPERIMENTAL EVALUATION



All results are consistent for LIME

4. CONCLUSIONS

The **LLM** is not a predictive model nor an explainer; it **acts as an interpretive semantic interface** over XAI outputs. **This work contributes to a evaluation approach for LLM-mediated explanations**, grounded in epistemic criteria such as fidelity and factuality. LLMs can also function as cognitive support tools, helping domain experts (e.g., drug designers) interpret and contextualize XAI explanations. **Future work** will extend the framework by integrating additional XAI methods, and by developing computational approaches for: robustness analysis, reliability assessment, and scalable evaluation of LLM–XAI pipelines.

Scan to view code and experimental proofs

References: **[1]** Yuan et al., IEEE TPAMI, 2022. **[2]** Rudin & Radin, HDSR, 2019. **[3]** Gilpin et al., IEEE DSAA, 2018. **[4]** Togo et al., JCIM, 2022. **[5]** Lombardo et al., CCJ., 2010. **[6]** Shapley, L. S., PUPP, 1953. **[7]** Ribeiro et al., KDD, 2016. **[8]** Mavrepis et al., arXiv, 2024. **[9]** *https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash*